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The Embodiment Abstraction: A Specification for the Coordination Layer between Vision-Language-Action Models and Robotic Embodiments

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0.1 Abstract

The physical-AI ecosystem of 2026 faces a coordination crisis. Nine credible Vision-Language-Action (VLA) foundation models and approximately thirty distinguishable robotic embodiments yield a combinatorial integration burden that, at an estimated median of \$300,000 per pair, imposes a fragmentation tax plausibly on the order of low single-digit billions of dollars annually and rising. Adjacent efforts — ROS 2, Open X-Embodiment, LeRobot, the single-model VLAs themselves, and NVIDIA's Isaac and GR00T stack — each address fragments of the problem but none occupies the specific structural position the crisis demands: a cross-vendor, semantically-typed, production-grade, neutrally-governed abstraction layer between intelligence and embodiment. We propose the Robot Foundation Layer (RFL), a three-layer specification consisting of a Skill ISA of fifty primitives and a compositional algebra, a canonical embodiment-agnostic action representation with a deterministic retargeting algorithm, and a Driver Interface compatible with ROS 2, all governed by five first principles: embodiment-agnostic, compositional, verifiable, provider-neutral, and forward-compatible. The network-effect architecture of RFL combines two-sided market structure with bilateral intelligence between models and embodiments, supermodular translation quality, and nine reinforcement loops that compound multiplicatively, yielding a defensibility ceiling we argue exceeds the historical analog of the ARM ISA. The specification is stewarded by a Linux Foundation subsidiary with binding constitutional commitments to neutrality, permissive licensing, scope discipline, transparency, and welfare primacy. We invite seven Founding Members — drawn from VLA providers, embodiment manufacturers, and research institutions — to ratify v0.1 in 2027 Q1, against a closing window we assess at approximately eighteen months.

0.2 Keywords

physical AI, vision-language-action models, robotic manipulation, foundation models, abstraction layers, two-sided markets, network effects, standards, ARM ISA, open governance

1 § 1. The Coordination Problem

The capability gap between what AI models can describe and what robotic systems can perform has, in the past three years, inverted. As recently as 2023, the binding constraint on physical AI was the model — robots could execute well-defined motions but could not generalize across tasks because the underlying intelligence was too brittle. By the second half of 2025 this constraint had moved. Vision-Language-Action (VLA) foundation models — $\pi 0$ from Physical Intelligence, Gemini Robotics from Google DeepMind, Helix from Figure, GR00T from NVIDIA, OpenVLA from Stanford / UC Berkeley / Toyota Research Institute, and a growing cohort of open-source successors — now generalize across object categories, manipulation primitives, and natural-language instructions in ways that, a decade ago, would have been classified as general AI in the manipulation domain.

The binding constraint has moved to the boundary between these models and the physical bodies they are supposed to inhabit. Every VLA must be re-integrated, by hand, against every

embodiment it is to control. Every embodiment vendor must repeat this integration with every VLA it wishes to support. The combinatorial cost has, in 2026, become the dominant drag on the field. This section frames the problem structurally, argues that the present window is the inflection at which the industry must coordinate on a shared abstraction or accept a generation of vertically integrated lock-in, and sets up the remainder of the paper.

1.1 1.1 The $N \times M$ Explosion

Two enumerations frame the problem.

On the model side, approximately nine VLA foundation model lines are credible candidates for industrial deployment in 2026 — credible meaning they have published checkpoints or commercial APIs, demonstrated multi-task generalization in third-party evaluations, and twelve-month forward maintenance commitments from an organization with public visibility. The set, in alphabetical order: Figure's Helix; Google DeepMind's Gemini Robotics; Hugging Face's SmolVLA and VLA-JEPA lines; NVIDIA's Isaac GR00T family; Physical Intelligence's $\pi 0$ family; Stanford / UC Berkeley / Toyota Research Institute's OpenVLA; Tsinghua's RDT-1B; and the Octo family from UC Berkeley. The count excludes academic checkpoints without industrial maintenance commitments and proprietary models known to be in development at major laboratories but not publicly disclosed.

On the embodiment side, the count is larger and growing faster. Restricting to manipulation-capable end effectors and arms with commercial or near-commercial availability as of mid-2026, the set includes humanoid platforms from Tesla, Figure, 1X, Sanctuary AI, Apptronik, Agility Robotics, Boston Dynamics, Unitree, and a cohort of Chinese entrants; multi-finger hands from Inspire Robotics, Wonik (Allegro Hand), CMU (LEAP Hand, open source), Shadow Robot Company, Schunk, Bridgestone (TETOTE pneumatic series), and roughly twenty research-grade hands maintained by academic labs; and conventional collaborative arms from Universal Robots, Franka Emika, Kuka, ABB, UFactory, Doosan, and Yaskawa. Conservatively deduplicated, this yields approximately thirty distinguishable embodiments in active 2026 deployment, with credible projections of fifty by end-of-2027 as the humanoid cohort matures.

The combinatorial implication is direct. Nine VLAs and thirty embodiments generate 270 candidate integration pairs. The realistically valuable subset is smaller — not every pair is commercially meaningful — but still exceeds one hundred pairs today and trends toward two hundred as both sides grow. No actor in the field is equipped to maintain anything approaching this number of integrations.

The current response, observable in 2026, is vertical lock-in: each VLA developer prioritizes a small number of embodiments, and each embodiment developer prioritizes a small number of VLAs. The market fragments into eight or ten vertically integrated stacks, each of which captures a fraction of the latent value because no stack achieves the cross-pollination that has driven every previous wave of platform progress in computing.

1.2 1.2 The Integration Cost

A single non-trivial integration — bringing a VLA from supporting one embodiment to supporting an additional one at production-grade reliability — consumes between two and six

engineer-months of senior robotics-AI talent. The lower bound applies when the new embodiment is structurally similar to one the VLA already supports. The upper bound applies when the embodiment introduces novel actuation modality (pneumatic versus tendon-driven), novel proprioceptive sensors (visuotactile versus force-only), or novel kinematic structure (a six-finger hand instead of a four-finger one).

At prevailing senior robotics-AI engineering wages in the dominant US labor market, this translates to a per-pair direct labor cost ranging from the low tens of thousands of dollars at the lower bound to the low hundreds of thousands at the upper bound, with similar ranges in Europe and Asia after currency adjustment. To this must be added three indirect costs:

1. Hardware procurement and access — acquiring and instrumenting at least one unit of the target embodiment for the duration of the integration window.
2. Data collection and labeling — teleoperation rigs, expert demonstrators, and curation pipelines to produce embodiment-specific demonstration data.
3. Certification and safety qualification — ISO 10218 / 13482 assessment when commercial deployment is the target.

The total per-pair cost is substantial, and the precise figure varies widely with task complexity, deployment target, and embodiment novelty. No public benchmark exists; the figures here are author estimates intended to convey scale rather than measurement.

1.3 1.3 What This Cost Suppresses

The aggregate paid integration cost across the industry is large enough to be macroscopic, but the more consequential cost is what does not get attempted. A research lab evaluating whether to test its new VLA on a structurally novel hand must trade six engineer-months against the project's other commitments; in most cases, the test does not happen. A startup considering a new robot form factor must accept that VLA support will lag the hardware by twelve to eighteen months. A potential industrial customer evaluating a robotic solution must price in not only the hardware and software they buy but the integration risk of pairing them.

This capability suppression is the largest component of the fragmentation cost and the hardest to measure. It does not appear in any firm's books because it consists of attempts not made. A defensible proxy is the gap between observed and counterfactual market growth in segments where integration cost is binding; this gap is directionally large but resists precise measurement, and we offer no point estimate. What matters for the argument that follows is the qualitative claim, which we take to be uncontroversial: the opportunity cost of integrations not attempted exceeds, by an order of magnitude or more, the direct cost of integrations made.

1.4 1.4 Why 2026 Is the Inflection Point

The combinatorial pressures described above have existed in some form since the field's inception. What makes 2026 distinctive — and what makes coordination on a shared abstraction layer urgent in this specific window — is the simultaneous occurrence of three structural inflections.

Inflection one: VLA models cross the threshold of cross-embodiment generality. Through 2024,

the dominant assumption was that a model trained on one embodiment would transfer poorly to another. The Open X-Embodiment results published by Google and thirty-three partner institutions in 2023, $\pi 0$'s multi-embodiment training in 2024, and Gemini Robotics's transfer results in 2025 falsified this assumption. By 2026 the question is no longer whether a single VLA can support many embodiments but rather how many embodiments will be available to support. The model side is hungrier than the embodiment side can feed.

Inflection two: embodiments diversify faster than they consolidate. The humanoid wave has not produced a single dominant form factor. Tesla Optimus's five-finger tendon design, Figure 03's twenty-four DOF hand, Sanctuary's hydraulic actuation, Bridgestone TETOTE's six-finger pneumatic structure, and the emerging Chinese cohort all represent distinct kinematic and actuation choices, each with a credible industrial sponsor. The expected consolidation toward a single anthropomorphic form has not occurred and shows no sign of occurring on a timeline shorter than a decade. The embodiment side is proliferating, not concentrating, at exactly the moment the model side is demanding it.

Inflection three: capital is committed at unprecedented scale, but coordination capital is absent. Through 2024 and 2025 the physical-AI sector raised disclosed venture financing at a scale measured in tens of billions of dollars, with humanoid robotics accounting for a substantial fraction. This capital has flowed almost entirely into vertical integration — VLA model training, specific robot designs, proprietary data collection — and not into the horizontal coordination layer that would multiply the value of the vertical investments. The field has solved the funding problem for individual stacks and produced a coordination problem in its place.

These three inflections compound. A field with stagnant model generality, consolidating embodiments, or limited capital would not need an abstraction layer urgently — bilateral integration would scale acceptably. A field with all three growing simultaneously, as in 2026, requires a layer that arbitrates the combinatorial growth or it will fragment into stacks that capture, in aggregate, a fraction of the available value.

The historical analog is instructive. The personal-computer industry of the early 1980s faced a structurally similar moment: software was becoming general enough to demand cross-hardware portability, hardware was diversifying rather than consolidating, and capital was abundant. The coordination layer that emerged — the IBM PC architecture, eventually inherited by the x86 ISA and the BIOS conventions that became the de facto standard — captured immense value precisely because it appeared in the brief window when the alternative (per-vendor verticalization) was just barely tolerable. A coordination layer proposed five years earlier would have lacked the demand pull; five years later would have arrived after the lock-in had ossified. The window was eighteen months, perhaps two years, and the right answer in that window shaped the industry for the next four decades.

1.5 1.5 The Question Posed

The diagnosis is precise. The physical-AI industry of 2026 is converging on a coordination failure: model generality is high, embodiment diversity is high, capital is high, and the integration interface between models and bodies is bespoke and bilateral. Vertical integration has emerged as the rational firm-level response to the absence of a coordination layer, and the re-

sulting balkanization is locking in the precise structure that long-run progress in physical AI cannot afford.

The remainder of this paper argues that the appropriate response is the construction of a neutral, semantically-typed abstraction layer — the Robot Foundation Layer (RFL) — that occupies the same structural position in physical AI that ARM occupies in silicon, that USB occupies in peripheral interconnects, and that POSIX occupies in operating systems. § 2 reviews why existing efforts (LeRobot, Open X-Embodiment, ROS 2, NVIDIA Isaac, and others) have each failed to occupy this position despite addressing adjacent concerns. § 3 develops the first principles from which the RFL design follows. § 4 presents the specification at version 0.1, including the interface contract for the deterministic retargeting algorithm that anchors the layer. § 5 places RFL in comparative context against ARM, CUDA, USB, ROS 2, and adjacent efforts. § 6 outlines the implementation roadmap. § 7 concludes.

The window is open. It will not remain so indefinitely.

2 § 2. Why Existing Approaches Fail to Occupy the Coordination Layer

The coordination crisis described in § 1 is not invisible to the field. Substantial engineering effort over the past five years has gone into systems that address adjacent fragments of the problem, and several have accumulated real momentum. Yet none has converged on the specific structural position that the crisis demands: a cross-vendor, semantically-typed, production-grade abstraction layer governed neutrally between Vision-Language-Action (VLA) models and robotic embodiments. This section examines the five most credible candidates — ROS 2, Open X-Embodiment, LeRobot, the single-model VLAs themselves, and NVIDIA's Isaac/GR00T stack — and argues that each is structurally precluded from occupying the position even on its natural roadmap. The argument is not that these projects are wrong or unimportant; they are correct and important within the problems they were built to solve. The argument is that none of those problems is the same as RFL's problem.

2.1 2.1 ROS 2: A Communication Layer Without Semantics

The Robot Operating System, in its second-generation form (ROS 2, built on the Data Distribution Service standard), is the closest thing the robotics field has to a universally adopted convention. Its action-server abstraction, its message-type system, and its publish-subscribe transport are deployed in thousands of research and commercial robots. If any existing system could expand upward into the coordination role RFL would occupy, ROS 2 is the *prima facie* candidate.

It cannot. ROS 2's design center is the low-level transport of typed messages between processes — “send a `geometry_msgs/Twist` to topic `/cmd_vel`” — and its action-server abstraction defines the shape of long-running operations (goal, feedback, result) but not their meaning. There is no canonical ROS 2 definition of what it means for a robot to “grasp a cable end and insert it into a connector”; there is only a convention for how such an action's interface might be structured if defined. The semantic gap between “I can transport bytes reliably” and “I can convey the

meaning of grasp across two different hands so that a single high-level model can drive both” is precisely the gap RFL must close.

Expanding ROS 2 to close this gap is not a roadmap question; it is an identity question. The current ROS 2 contract — backward-compatible, transport-focused, runtime-agnostic — would have to be broken to introduce action semantics, and breaking it would alienate the deployed base on which its credibility rests. The right move for ROS 2 is to remain the transport layer beneath RFL, not to attempt to subsume it. § 4 of this paper makes the integration with ROS 2 explicit: RFL's Driver Interface is designed to expose itself as a set of ROS 2 action servers, so that the existing transport investment is preserved.

2.2 2.2 Open X-Embodiment: A Dataset Without a Standard, a Stewardship Without Neutrality

Open X-Embodiment, the Google-led initiative joined by thirty-three partner institutions in 2023, has done more than any other single effort to establish the empirical foundation for cross-embodiment generalization. The pooled dataset — over a million teleoperated trajectories spanning twenty-two embodiments — supplied the training corpus on which RT-2, $\pi 0$, and subsequent VLAs demonstrated their cross-platform transfer results. The intellectual contribution is substantial.

The structural contribution, however, stops at the dataset boundary. Open X-Embodiment defines a data format (largely the RLDS format from Google's prior data work) but not a runtime standard. It tells researchers how to label demonstration trajectories from a new embodiment; it does not tell a production deployment how to route a high-level instruction from a VLA to that embodiment at runtime. A VLA trained on Open X-Embodiment data still requires bespoke integration to control a new embodiment in deployment — the dataset standard does not bridge to a control standard.

The second blocker is stewardship. Open X-Embodiment is housed at Google, and Google maintains its own VLA model line (Gemini Robotics) that competes commercially with other VLA providers. This does not impugn the integrity of the academic collaboration, but it does foreclose the trajectory by which Open X-Embodiment could evolve into the neutral coordination layer. The OEM that adopts an abstraction standard housed at a hyperscaler with its own VLA product is making a different commercial bet than the OEM that adopts a standard housed at a neutral foundation. Cisco and 3Com were never going to adopt a networking standard owned by IBM; the same dynamic applies here.

2.3 2.3 LeRobot: Strong Momentum, Wrong Center of Gravity

LeRobot, the open-source robotics framework that Hugging Face introduced in 2024 and has aggressively scaled through 2025–2026, is the candidate most often cited in conversations about coordination-layer convergence. It has the right cultural disposition (open source, community-driven), the right developer-experience instincts inherited from the Hugging Face transformers ecosystem, and the most rapidly accumulating contributor base of any project in the field. If RFL is to encounter a serious competitor for the structural position it seeks, the realistic candidate is LeRobot.

The reason LeRobot does not yet — and structurally may not — occupy the RFL position lies in two divergent design pressures. First, LeRobot's center of gravity is the ML-researcher workflow: train a policy, evaluate it on a benchmark, share the checkpoint. This is the right design center for accelerating research, and LeRobot has done it superbly. It is not the right design center for production deployment in industrial environments, which require real-time safety guarantees, certifiability against ISO 10218/13482, deterministic latency, and integration with existing PLC and field-bus infrastructure. The features that make LeRobot a great research framework — Python-first orientation, flexibility, rapid iteration — work against the deterministic, statically analyzable, safety-certifiable substrate that industrial deployment demands.

Second, Hugging Face's commercial trajectory tends toward vertical platform consolidation rather than standards stewardship. The Hugging Face Hub is itself a platform with proprietary monetization (Spaces, Inference Endpoints, Enterprise tier), and the natural commercial path for LeRobot is to become the gateway to Hugging Face's robotics services. This is a legitimate business model, but it is structurally incompatible with the neutral-foundation governance that the coordination layer requires. An OEM that builds on LeRobot is building on infrastructure whose operator competes commercially with adjacent layers — exactly the trap that ARM avoided by separating itself from Acorn.

A possible counter is that Hugging Face could spin LeRobot out into a neutral foundation analogous to the Linux Foundation. We treat this as the most credible scenario in which the RFL position is contested rather than empty, and § 9's call to action explicitly invites collaboration with LeRobot stewards under such a structure. As of mid-2026 no such restructuring has been signaled.

2.4 2.4 Single-Model VLAs: Applications That Cannot Be Platforms

A more provocative claim than any of the above is that the leading VLA models themselves — $\pi 0$ from Physical Intelligence, OpenVLA from Stanford, GR00T N1 from NVIDIA — cannot become the coordination layer regardless of how technically capable they grow. The reasoning is structural, not technical, and it applies even to the most generous interpretation of any of these models' future capability.

A VLA is an application: an artifact that takes perception and instruction in and produces actions out. It competes, in the market, with other applications that do the same thing. Two VLAs competing for the same customer's deployment cannot simultaneously serve as the neutral substrate on which that customer's robot is built. Physical Intelligence's decision to open-source the $\pi 0$ weights does not change this; the weights are now free, but the integration interface — what counts as a valid $\pi 0$ input, what semantics its action outputs carry, how it expects embodiment metadata — remains controlled by Physical Intelligence and follows Physical Intelligence's commercial roadmap. An OEM that builds its embodiment-side software against $\pi 0$'s specific interface is taking a bet on Physical Intelligence, not on a standard.

The dynamic is identical to the early 1990s in operating systems: Sun Microsystems' Solaris was excellent, IBM's AIX was excellent, HP-UX was excellent, but none of them could serve as the substrate for a multi-vendor application ecosystem because each was a commercial product of a vendor that competed for the same customers. The substrate that did emerge — POSIX

as a specification, eventually Linux as an implementation — was created precisely because no incumbent could occupy the structural position. The physical-AI field is at the same juncture with respect to VLAs.

The implication is not that $\pi 0$, OpenVLA, and GR00T are unimportant; they are the applications that will run on top of RFL. The implication is that none of them, on its present trajectory, can become RFL.

2.5 2.5 NVIDIA Isaac and GR00T: The CUDA-Adjacent Trap

NVIDIA's Isaac platform, paired with the GR00T N1 humanoid foundation model and the Cosmos world model, is the most technically capable adjacent stack on offer in 2026. It bundles simulation (Isaac Sim, Isaac Lab), perception, motion planning, foundation models, and deployment tooling into the most integrated end-to-end physical-AI offering currently available. By raw capability it is, plausibly, the strongest existing candidate.

By neutrality it is the weakest. Isaac's value proposition is inseparable from CUDA: the simulation runs on NVIDIA GPUs, the foundation models train on NVIDIA hardware, the inference deploys on NVIDIA edge devices (Jetson, Thor). This is not a deficiency in the engineering; it is the engineering. NVIDIA built Isaac to sell NVIDIA silicon, and the architecture reflects that goal at every layer. The result is that any embodiment vendor or VLA provider that builds against Isaac is, structurally, accepting a CUDA dependency in perpetuity.

A coordination layer cannot be CUDA-dependent because half the relevant ecosystem is structurally incentivized to escape CUDA: Apple's Neural Engine, Google's TPU, Tesla's Dojo, Qualcomm's Hexagon, and an emerging cohort of ARM-based and RISC-V-based edge accelerators all need a path to physical-AI deployment that does not route through NVIDIA's tooling. Even within NVIDIA's customer base, the AI-chip market's recent volatility has made customers cautious about deepening any single-vendor dependency. An abstraction layer whose adoption signals long-term lock-in to NVIDIA is, for half its potential adopters, worse than no abstraction layer at all.

The case is parallel to OpenGL versus Direct3D in the late 1990s: Direct3D was technically superior in many releases, but its Microsoft dependency drove the cross-platform graphics market to consolidate around the neutral alternative. RFL's relationship to Isaac is the same: cooperative where possible (RFL implementations should run on Isaac Sim as one of several supported simulators), structurally distinct in governance.

2.6 2.6 The White Space, Mapped

Set the five existing efforts against the coordination layer's requirements and the gap becomes geometrically visible.

Requirement	ROS 2	Open X-E	LeRobot	Single-model VLAs	NVIDIA Isaac	RFL
Cross-vendor VLA support	n/a	n/a	partial	by definition no	partial	✓

Requirement	ROS 2	Open X-E	LeRobot	Single-model VLAs	NVIDIA Isaac	RFL
Cross-vendor embodiment support	partial	partial	partial	partial	partial	✓
Semantically-typed actions (not just messages)	×	×	partial	proprietary	proprietary	✓
Production-grade (real-time, certifiable)	✓	×	weak	varies	✓	✓
Neutral governance	✓	weak (Google)	weak (HF)	by definition no	×	✓
Standalone runtime value (not just data/training)	✓	×	✓	✓	✓	✓

Each existing effort satisfies a non-trivial subset of the requirements. None satisfies the full set. The conjunction of semantically typed, cross-vendor on both sides, production grade, and neutrally governed defines an empty point in the existing landscape. That point is the position RFL is designed to occupy.

A subtler observation, visible only when the five efforts are considered together, is that the field's coordination has been over-attempted at the wrong layers rather than under-attempted in the aggregate. There is no shortage of will to coordinate; there is a shortage of will to coordinate at the specific layer where coordination would have the largest network-effect leverage. ROS 2 coordinated transport; Open X-Embodiment coordinated training data; NVIDIA coordinated tooling; LeRobot coordinated research workflow. Each of these coordinations is real and useful. None of them is a coordination of semantically meaningful action between competing model providers and competing embodiment providers. That coordination is what RFL proposes.

2.7 The Implication for Design

Two design implications follow directly from the diagnostic above and govern the choices in § 3 and § 4.

First, RFL must be additive to existing infrastructure rather than replacement. ROS 2 is the right transport; Isaac Sim is a fine simulator; Open X-Embodiment data is excellent training material; LeRobot's developer experience is worth borrowing where compatible. The argument of this paper is not that the existing stack is wrong, but that it is incomplete, and that completing it

requires a single new piece at a specific layer. The RFL specification is designed to compose with each of the above rather than compete with them.

Second, the governance question is as load-bearing as the technical question. The reason every existing effort fails to occupy the coordination position is, in three of five cases, governance rather than technology. A technically perfect specification housed under the wrong steward will not be adopted; a technically average specification housed under the right steward will be. § 7 of this paper develops the foundation governance structure in detail, but the design pressure is established here: the steward must be a neutral foundation, the spec must be permissively licensed in perpetuity, and the founding members must be drawn from genuinely competitive corners of both sides of the market.

The remainder of the paper takes these design pressures as constraints and develops the specification, network-effect architecture, and governance structure that satisfy them.

2.8 Open editorial notes

- Word count: $\approx$ 2,100 words (target 1,800–2,200 ✓)
- Tone calibration: Each subsection acknowledges genuine contribution before identifying the structural blocker. This is important; the founding consortium and foundation steward roles will, in practice, require warm relations with the maintainers of every project critiqued here. No strawmanning.
- Verification flags before final:
 - Open X-Embodiment dataset size (1M+ trajectories, 22 embodiments) — confirm with current Open X-E paper version
 - LeRobot contributor count and Hub statistics — confirm latest
 - NVIDIA Isaac/Cosmos/GR00T product line — confirm current taxonomy and naming
 - Confirm Hugging Face has not announced a LeRobot foundation spin-out as of publication
- Figures to commission:
 - 2.1: Comparison matrix (the table in § 2.6) rendered as a visual capability grid with green/yellow/red cells
 - (optional) 2.2: Layer diagram showing where each existing effort sits in the physical-AI stack and the gap that RFL fills
- Cross-reference dependencies:
 - § 3 inherits the “neutral foundation” and “additive not replacement” design pressures
 - § 4 inherits the requirement that RFL expose itself through ROS 2 action server semantics
 - § 7 inherits the governance-as-load-bearing argument
 - § 9 inherits the explicit collaboration invitation to LeRobot stewards

2.9 Coverage check against outline (STRATEGY.md § 15.2)

Outline item	Section	Status
2.1 ROS 2	§ 2.1	✓
2.2 Open X-Embodiment	§ 2.2	✓
2.3 LeRobot (HF)	§ 2.3	✓
2.4 OpenVLA / $\pi 0$	§ 2.4 (broadened to include GR00T)	✓
2.5 NVIDIA Isaac/GR00T	§ 2.5	✓
2.6 The white space	§ 2.6	✓
(added) Design implications	§ 2.7	✓

2.10 Revision history

Version	Date	Notes
v1.0	2026-05-29	First English draft. Followed STRATEGY.md § 15.2 outline directly.

3 § 3. RFL: First Principles

A specification proposed without first principles is a specification that drifts under commercial pressure. Every layer that has achieved durable cross-vendor adoption — POSIX, the C ABI, the ARM ISA, USB, HTTP — was anchored by a small set of design commitments that were stated before the technical details and that constrained the technical details. RFL adopts five such commitments. They are derived from the constraints established in §§ 1–2 (the combinatorial crisis on one side; the structural failures of adjacent efforts on the other), and every choice in § 4’s specification can be traced back to them.

This section states the five principles, draws the structural isomorphism with the ARM ISA on which the RFL design pattern is modeled, and identifies the dimensions on which RFL must deliberately diverge from ARM because the physical-AI substrate differs from silicon in ways that compatibility intuition does not capture.

3.1 3.1 The Methodological Commitment

Before stating the principles, a methodological commitment: principles, not features. A design principle is a constraint that survives commercial pressure because adherents to the standard refuse to violate it even when individual short-term gains would be available from doing so. A feature is a capability that exists because someone wanted it. Principles can be evaluated against future situations the original drafters did not anticipate; features cannot. The history of every coordination-layer success in computing shows the same pattern: a small set of principles, vigorously defended, outperforms a large set of features, opportunistically accumulated.

The five principles below are the smallest set that, in our analysis, jointly forecloses each known failure mode of past abstraction-layer attempts. We have considered and rejected larger prin-

principle sets (the early POSIX standardization process accumulated more than a dozen guiding tenets and slowed correspondingly) and smaller principle sets (a three-principle articulation we initially favored failed to constrain the governance dimension and would have permitted vendor capture). Five is, on our analysis, the parsimonious count.

3.2 The Five Principles

3.2.1 Principle 1: Embodiment-Agnostic

The specification must avoid encoding the design choices of any particular embodiment. Action semantics must be expressible across tendon-driven, pneumatic, hydraulic, and as-yet-unknown actuation modalities; across four-finger, five-finger, six-finger, and parallel-jaw end effectors; across anthropomorphic and non-anthropomorphic kinematic structures.

This is the foundational constraint and the easiest one to violate. Almost every existing robotics convention is, on examination, quietly encoded with the assumptions of a dominant embodiment class. The Feix grasp taxonomy assumes the human hand; ROS 2's `geometry_msgs` family assumes Cartesian-end-effector control; the dominant teleoperation interfaces assume two-handed bimanual operation. RFL must not bake in any such assumption.

The operational test: a draft of the specification must be reviewable by someone whose only experience of robotics is a non-anthropomorphic platform — say, a pneumatic six-finger hand or a soft-bodied actuator — and they must find the semantics expressible in their world without translation through anthropomorphic intuition. If they cannot, the spec has smuggled in an embodiment assumption and must be rewritten.

3.2.2 Principle 2: Compositional

Skills are atomic primitives that compose, under a defined algebra, into higher-level behaviors. The specification defines the composition rules, not merely the primitive set.

A skill ISA that is only a list of primitives is a skill library, not a skill language. The difference matters because the value of an abstraction layer grows multiplicatively, not linearly, with the expressiveness of its composition rules. ARM's enduring success rests on its instruction set being composable into procedures, libraries, and operating systems through a small set of calling conventions and addressing modes; the primitives matter less than the composition rules built atop them.

RFL's Skill ISA defines, in addition to the primitive operations, the rules by which sequential, conditional, and concurrent composition is expressed, the way pre- and post-conditions of composed skills are derived from their components, and the algebra by which the safety envelope of a composition is inherited from its components. § 4.4's worked example exercises these composition rules in a single multi-step manipulation; § 4.1 enumerates the primitive set.

3.2.3 Principle 3: Verifiable

Every skill carries pre-conditions, post-conditions, and a safety envelope expressible in a machine-checkable form. Composition rules preserve verifiability.

The verification principle distinguishes RFL from an API. An API dispatches function calls; a verifiable abstraction layer additionally promises that, if the pre-conditions hold and the skill executes to its post-conditions, certain safety-relevant invariants will hold throughout the execution. This promise is what makes the Certification reinforcement loop (§ 5.4 L4) and the Insurance reinforcement loop (§ 5.4 L8) possible. Without verifiability, RFL cannot interoperate with the regulatory and actuarial institutions that physical-world deployment ultimately requires.

The verification principle imposes design discipline on the primitive set. A primitive whose safety envelope cannot be expressed compactly is rejected, regardless of how useful the primitive would otherwise be. This is a real cost; some genuinely useful operations (notably those involving deformable-object manipulation) will lie at the edge of expressibility and require iterative spec refinement. We accept the cost because the alternative — admitting unverifiable primitives — corrupts the certifiability of every composition that includes them.

3.2.4 Principle 4: Provider-Neutral

No vendor's preferred way of doing things has special status in the specification. The specification's evolution is governed by procedures that treat Founding Members and late entrants identically once admitted.

Provider neutrality is governance expressed at the level of the spec. It is the technical instantiation of the foundation governance principles developed in § 7. The principle has two operational tests: first, an outside reviewer should be unable to identify which Founding Member proposed any given primitive by inspecting the spec; second, the spec's evolution procedure (voting, comment periods, breaking-change moratoria) should produce the same outcome regardless of which member proposes the change.

This principle is what foreclosed the otherwise tempting design choice of letting the most technically capable Founding Member contribute a head-start spec and have other members ratify it. The technical benefits of such an approach (faster spec maturity, fewer compromises) are real, but the perception of vendor capture — and the actual erosion of cross-vendor adoption that perception would produce — is fatal. The compromise we adopt instead is that the v0.1 reference specification is drafted by the RFL Foundation's neutral spec-editor staff, with proposals from any Founding Member receiving equal procedural treatment.

3.2.5 Principle 5: Forward-Compatible

The specification is designed to extend without breaking. Future VLA capabilities and future embodiment classes must be expressible in extensions; the v1.x line must never deprecate v1.0 functionality.

The forward-compatibility principle is the discipline that distinguishes a standard from a moving target. Backward compatibility within a major version is not optional. New primitives, new composition rules, and new metadata fields enter the spec through additive extensions, organized into a versioned extension registry analogous to RISC-V's modular extensions or x86's CPUID feature bits. A VLA that targets RFL v1.0 in 2027 must continue to work against

RFL-compliant embodiments shipped in 2032; an embodiment certified against RFL v1.0 must continue to receive VLA support indefinitely.

This principle imposes a non-trivial intellectual cost on the v0.1 design: the drafters must anticipate which extensions are foreseeable and structure the core spec so that those extensions can be added without disturbing existing semantics. We have used the design heuristic of capability discovery (every RFL-compliant component exposes its capabilities through a structured manifest) and graceful degradation (a VLA that requires a capability the embodiment lacks must receive a clear failure mode, not undefined behavior). § 4.3 develops both mechanisms in detail.

3.3 Isomorphism with the ARM ISA

The five principles, taken together, place RFL in structural correspondence with the ARM Instruction Set Architecture. The correspondence is not metaphorical; it is point-for-point.

RFL element	ARM ISA element
Skill ISA primitives	Instruction set primitives
Skill composition rules	Calling conventions, ABI
Pre/post-condition contracts	Architectural invariants (e.g., register state preservation)
Driver Interface	Memory model, peripheral I/O conventions
Capability discovery manifest	CPUID-equivalent feature reporting
Extension registry	ARM architecture profiles (A/R/M, plus extensions)
Foundation governance	ARM Holdings as steward
Compliance test suite	ARM architecture validation suite

The correspondence is generative, not accidental. The first principles in § 3.2 were derived in part by asking, for each ARM ISA design choice that proved durable: what was the structural reason that choice succeeded, and what is the physical-AI analog of that reason? Embodiment-agnosticism corresponds to ARM's silicon-agnosticism (the ISA does not specify how the chip is fabricated); compositionality corresponds to ARM's procedure-and-library compositionality (the ISA defines how procedures call each other); provider-neutrality corresponds to ARM Holdings' decision not to manufacture chips itself; forward-compatibility corresponds to ARM's discipline of architectural-version stability across decades.

The historical evidence is strong. ARM's enduring position in silicon emerged precisely from disciplined adherence to a small set of principles that resisted commercial pressure to deviate. Two specific moments are illustrative: ARM's refusal to acquire chip-fabrication capacity in the early 2000s (which would have grown short-term revenue but destroyed the neutrality that anchored vendor adoption); and ARM's decision in the late 2010s to extend the architecture (ARMv8, ARMv9) through additive features rather than breaking changes (which preserved the multi-decade application binary base). RFL's principles 4 and 5 are direct descendants of those decisions.

3.4 3.4 Where RFL Must Diverge from ARM

The isomorphism, however, is not complete. The physical-AI substrate differs from silicon along four dimensions that the ARM design did not need to address and that RFL cannot inherit unchanged.

Safety as architectural concern. ARM's instructions do not physically harm. A buggy ARM program can hang the machine, corrupt data, or leak information, but it cannot crush a human hand. RFL's primitives can. The verifiability principle (Principle 3) and the safety envelope mechanism it requires have no ARM analog and must be designed from first principles for physical action. § 4.2's translation layer and § 4.4's worked example show the safety envelope's structure.

Two-sided market urgency. ARM emerged into a market where chip vendors and OS providers were not racing each other to vertical integration. The silicon ecosystem had time to consolidate around the ISA. RFL faces a market in which vertical integration (Figure–Helix, Tesla–Optimus, NVIDIA–GR00T) is already underway and accelerating; the window for a horizontal coordination layer is narrower than ARM's was, and the founding-member commitments must be secured before vertical lock-in ossifies. This urgency shapes the cold-start strategy in § 5.3 and the consortium recruitment plan in § 9.

Data gravity as a moat input. Silicon does not generate data that improves silicon. Physical AI does: every executed action becomes training data for the next generation of translation models. RFL's design must therefore include the data-flow architecture that captures and shares this learning across the ecosystem (the L1 Data reinforcement loop in § 5.4) without violating the privacy or commercial interests of the participating Founding Members. § 4.5's conformance test suite and § 7's data-sharing governance both descend from this concern.

Multi-jurisdictional regulatory exposure. ARM faced one significant regulatory question (export controls on advanced cores). RFL faces a dense thicket: ISO 10218 and 13482 on industrial-robot safety, the emerging EU AI Act provisions on autonomous systems, sector-specific medical-device regulation for healthcare embodiments, and a patchwork of national rules on AI accountability. The forward-compatibility principle (Principle 5) acquires a regulatory dimension that ARM never had: the spec must be extensible not only to absorb future capabilities but to absorb future regulatory frameworks without architectural disruption.

Each of these divergences makes RFL harder to design than ARM was. None of them invalidates the structural correspondence. The five principles remain the spine; the divergences are constraints on how the spine is fleshed out.

3.5 3.5 How the Principles Constrain the Specification

The principles are not aspirational; they are operational. § 4's specification is constructed under their constraint in the following specific ways.

The Skill ISA primitive set is restricted to operations whose pre/post-conditions and safety envelopes are expressible in the verification language (Principle 3) and whose semantics are stable across at least three structurally different embodiments in the v0.1 reference implementations (Principle 1). The Translation Layer's canonical action space is parameterized along axes that ad-

mit any kinematic and actuation modality currently known to be commercially feasible, rather than along axes that are convenient for a specific class of robots (Principle 1). The Driver Interface contract is designed so that any robot whose firmware exposes the contract can host any RFL-compliant VLA, without negotiation specific to that VLA (Principle 4). Every primitive carries a version stamp from its first appearance, and the v1.x evolution policy explicitly forbids retracting a stamped primitive (Principle 5). The composition rules are defined such that the safety envelope of a composed skill is computable from the safety envelopes of its components, enabling certification at composition granularity rather than requiring per-composition reverification (Principles 2 and 3).

These constraints make § 4 longer and more conservative than a feature-driven specification would be. They are the deliberate cost of building a layer that can survive the commercial pressure it will encounter, in the same way that ARM's disciplined architectural conservatism imposed costs in 1985 that paid off across the four decades that followed.

3.6 Open editorial notes

- Word count: $\approx 2,150$ words (target 1,800–2,200 ✓)
- Tone: deliberately principle-statement-with-rationale, modeled on the structure of the POSIX foundational documents and the early ARM Architecture Reference Manual prefaces. Not a manifesto.
- Verification flags before final:
 - “ARMv8 and ARMv9 extended through additive features rather than breaking changes” — directionally correct, but confirm with ARM's official architecture history before publication; ARMv8-A $\rightarrow$ ARMv9-A includes some changes worth honest characterization
 - “POSIX standardization process accumulated more than a dozen guiding tenets” — soft claim, confirm with IEEE 1003 history
 - Cross-reference §§ 4.1–4.5 to ensure each principle's spec embodiment is actually delivered in § 4
- Figures to commission:
 - 3.1: Five-principles diagram (pentagon with rationale on each vertex)
 - 3.2: RFL $\leftrightarrow$ ARM ISA isomorphism table (the table in § 3.3) rendered as a side-by-side correspondence diagram
 - (optional) 3.3: Four divergences from ARM as a 2 $\times$ 2 (silicon vs physical world $\times$ technical vs governance)
- Cross-reference dependencies:
 - § 4 inherits every principle as a binding design constraint
 - § 5.8 already inherits the five principles in the form of North Star principles for foundation governance
 - § 7 inherits the provider-neutrality principle as the basis for voting structure
 - § 8 inherits the ARM isomorphism for the comparative analysis

3.7 Coverage check against outline (STRATEGY.md § 15.2)

Outline item	Section	Status
3.1 □□□□ 5 □	§ 3.2 (each principle in own subsection)	✓
3.2 ARM □□□□□□	§ 3.3 (table)	✓
3.3 ARM □□□□	§ 3.4 (four divergences)	✓
(added) Methodological commitment	§ 3.1	✓
(added) Operational implication for spec	§ 3.5	✓

3.8 Revision history

Version	Date	Notes
v1.0	2026-05-29	First English draft. Followed STRATEGY.md § 15.2 outline directly.

4 § 4. The RFL Specification v0.1

This section presents the Robot Foundation Layer specification at version 0.1. It is deliberately narrow in scope: it specifies the abstraction layer between Vision-Language-Action (VLA) intelligence and robotic embodiment, and nothing else. Layers above the spec (task planning, world modeling) and below it (real-time motor control, sensor drivers) are explicitly out of scope, in accordance with the Layer Discipline principle (§ 5.8, derived from Principle 4). The spec defines three layers: a Skill Layer providing semantically typed action primitives and their composition algebra; a Translation Layer providing a canonical, embodiment-agnostic action representation; and a Driver Interface defining the contract between the Translation Layer and embodiment-specific firmware.

The spec is presented at a level of detail sufficient to fix the architectural commitments and to enable competent implementers to begin work. Full implementation detail — every type signature, every JSON schema, every error code — is deferred to the companion GitHub repository, where the spec lives as a versioned document subject to the evolution procedure described in § 7.

4.1 4.1 The Skill Layer

The Skill Layer defines fifty base primitives organized into seven categories, plus the compositional algebra that combines them into multi-step manipulations. The primitive count is itself a design commitment: too few primitives forces VLAs to express common manipulations through awkward compositions, while too many primitives encourages embodiment-specific encoding (Principle 1). Fifty is the count at which, in our v0.1 design review with the prospective Founding Consortium, every commonly cited industrial manipulation could be expressed in three or fewer composition steps, with no primitive applying to only a single embodiment class.

The categories and representative primitives:

Category	Count	Representative primitives
Grasping	10	grasp_power, grasp_precision, grasp_pinch, grasp_lateral, grasp_hook, grasp_cylindrical, release, regrasp, transfer_grasp, adjust_grip
Translation & placement	8	place, push, pull, slide, drag, stack, retract, advance
Rotation & orientation	6	rotate, twist, screw_in, screw_out, reorient, align
Insertion & connection	8	insert, seat, mate, snap, latch, clip, lock, plug
Continuous manipulation	8	pour, wipe, fold, unfold, smooth, crease, drape, wrap
Inspection & sensing	6	probe, palpate, scan, measure_force, measure_compliance, verify_state
Multi-effector coordination	4	bimanual_handoff, bimanual_hold_and_act, multi_finger_regrasp, cooperative_carry

Each primitive is specified through five fields: a type signature (typed inputs and outputs), pre-conditions (environmental and state predicates that must hold before invocation), post-conditions (predicates that hold upon successful completion), a safety envelope (force bounds, motion bounds, and abort triggers that hold throughout execution), and a capability requirement (the embodiment capabilities, declared through the Driver Interface, that the primitive requires for execution).

As a concrete example, the `grasp_pinch` primitive is specified as follows:

primitive `grasp_pinch`:

inputs:

```
target:      Object3D          # geometry + pose + material descriptor
force_budget: Force[N]        # default 5.0 N, max 50.0 N
grasp_pose:  Pose6D | "auto"   # default "auto" (planner-derived)
```

outputs:

```
result:      {success, slip, force_exceeded, unreachable}
final_pose:  Pose6D           # actual grasp pose achieved
telemetry:   GraspTelemetry   # tactile + force trace
```

preconditions:

```
- target.is_visible_to(embodiment.perception)
- target.estimated_mass <= embodiment.capabilities.payload_grasp_pinch
```

```

- embodiment.state.end_effector_free
postconditions:
- if result == success: object_held(target, mode=pinch)
- if result == slip:    embodiment.state.end_effector_free
safety_envelope:
- peak_force <= force_budget * 1.2          # 20% transient margin
- peak_velocity <= embodiment.limits.v_grasp
- abort_if: target.crush_indicator > 0.8
capability_requirement:
- {pinch_grasp: required}
- {tactile_sensing: preferred, level: "binary_contact" or higher}

```

The specification structure is uniform across all fifty primitives, which is what enables the verifiability principle (§ 3.2 Principle 3): a tool can mechanically check that every primitive in a composition has its pre-conditions established by predecessors and that the composition's safety envelope is the proper composition of the individual envelopes.

Compositional algebra. The Skill Layer defines five composition operators, sufficient to express the manipulation workflows we examined across the v0.1 design review.

Operator	Syntax	Semantics
Sequential	$A ; B$	Execute A to success, then execute B; if A fails, composition fails
Conditional	$A ? B : C$	Execute predicate A; on true execute B, on false execute C
Concurrent	$A \parallel B$	Execute A and B simultaneously; safety envelope is union; composition succeeds when both succeed
Reactive	$A \text{ until trigger}$	Execute A until trigger fires, then halt A cleanly
Bounded retry	$\text{try } A \text{ retries } n$	Execute A; on failure retry up to n times before failing the composition

Composition rules preserve verifiability. The pre-condition of a composition is the pre-condition of its first-executed component; the post-condition is the post-condition of the last component along the executed path; the safety envelope is the union of the components' envelopes for $;$ and $?$, and the strict intersection for $\parallel$ (any envelope violation by either concurrent action aborts both). Bounded retry inherits the envelope of A unchanged but admits multiple attempts within it.

A subtlety worth noting: the Concurrent operator's strict-intersection safety semantics is deliberately conservative, because two concurrent actions whose envelopes individually permit certain forces may, when executed together, produce force interactions that neither envelope

anticipated. The conservative choice forecloses a class of compositions that would, in some embodiments, be safe; we accept the cost as the price of mechanical verifiability.

4.2 The Translation Layer

The Translation Layer converts Skill Layer invocations into a canonical, embodiment-agnostic action representation that the Driver Interface can route to embodiment-specific firmware. It is the spec's most architecturally novel layer, occupying the position that ROS 2 does not (semantic translation, not just message transport) and that single-model VLAs cannot (cross-embodiment routing, not just internal model dispatch).

The canonical action representation is a tuple:

```
CanonicalAction = (  
    pose_trajectory:      SE(3) sequence with timestamps,  
    force_trajectory:     Force[6] sequence aligned with pose_trajectory,  
    tactile_target:       TactileManifold | None,  
    proprioceptive:       JointConfig | None,  
    capability_flags:     set of CapabilityFlag,  
    confidence:           DistributionOverSuccess,  
    safety_envelope:      SafetyEnvelope (inherited from Skill Layer)  
)
```

Five aspects of this representation are load-bearing.

First, the pose and force trajectories are expressed in the task frame, not in any embodiment-specific frame. The Driver Interface's calibration handshake (§ 4.3) is responsible for mapping task-frame coordinates into the embodiment's native frame at execution time. This indirection is what allows the same canonical action to drive radically different embodiments without re-targeting at the VLA layer.

Second, the tactile target is expressed in the TactileManifold abstraction — a topological description of contact patterns rather than a specific tactile-sensor output format. An RFL-compliant embodiment whose tactile sensors produce four-channel binary contact data can describe its observations in TactileManifold terms; an embodiment with high-resolution GelSight-class visuotactile output can describe its richer observations in the same terms with higher resolution. The abstraction is leaky in a controlled way: VLAs can request a minimum manifold resolution through capability flags, and the Translation Layer's capability negotiation will route the action to embodiments that meet the requirement (Principle 5: graceful degradation).

Third, the proprioceptive field is optional. Many Skill Layer primitives — particularly grasping and continuous manipulation — do not require joint-level specification, because the Translation Layer can derive a kinematically feasible joint trajectory from the task-frame trajectory and the embodiment descriptor. The field is present for the cases where the VLA has a specific reason to control joint configuration (avoiding obstacles, matching a learned policy's training distribution).

Fourth, the confidence field carries a distribution over outcomes rather than a point estimate. This is the input the Driver Interface needs to set appropriate safety margins: a high-confidence

action can execute with aggressive timing, a low-confidence action triggers conservative timing and richer telemetry capture.

Fifth, the safety envelope is inherited unchanged from the Skill Layer composition. The Translation Layer is responsible for ensuring that the canonical action it produces will not, when executed, violate the envelope; if no kinematically feasible trajectory exists within the envelope, the Translation Layer returns a failure to the Skill Layer rather than relaxing the envelope.

Retargeting interface contract. The mapping from a Skill Layer invocation to a canonical action depends on the target embodiment. The Translation Layer is parameterized by an embodiment descriptor (loaded from the Driver Interface at session initialization). The v0.1 specification fixes the interface contract for the retargeting function and defers the algorithmic implementation to v1.0 (2027 Q4 target). The interface is:

```
retarget : (Skill, EmbodimentDescriptor, EnvironmentState)
          → CanonicalActionSequence ∪ {⊥_safety, ⊥_unsupported}
```

Any conformant implementation must satisfy five binding invariants:

ID	Invariant	What it requires
I1	Determinism	retarget is a pure function; identical inputs produce byte-identical output across all conformant implementations on all platforms. No reliance on system clock, RNG, or thread scheduling permitted in the produced canonical-action bytes.
I2	Embodiment-respect	Every action in the output is executable by the embodiment described by the input descriptor under its declared kinematic and actuation envelope. The Translation Layer never emits an action the embodiment cannot perform.
I3	Composability	$\text{retarget}(\text{sequence}(s_1, s_2), e, \text{env}) = \text{concat}(\text{retarget}(s_1, e, \text{env}), \text{retarget}(s_2, e, \text{env}'))$ where env' is the environment state after s_1 executes. Analogous distributivity over parallel, repeat, and branch constructors. The Skill Layer's compositional algebra survives translation.

ID	Invariant	What it requires
I4	Failure-mode preservation	If <code>retarget</code> cannot satisfy the skill's safety envelope under the target embodiment, it returns <code>⊥_safety</code> explicitly rather than emitting a degraded or partially-safe action sequence. Silent envelope relaxation is non-conformant.
I5	Extension-namespace isolation	Extension primitives (<code>ext.<namespace>.<name></code>) flow through <code>retarget</code> unchanged when the embodiment descriptor declares support; <code>⊥_unsupported</code> is returned otherwise. The core <code>retargeter</code> does not interpret extension semantics.

The conformance suite (§ 4.5) defines two relevant test classes against these invariants:

- Class 2: Translation Layer determinism — for a fixed (s, e, env) triple, observed output across runs and platforms must be byte-identical. This class is testable against any candidate implementation, not only the eventual v1.0 reference, and is therefore the earliest layer of the spec to which conformance can be evaluated.
- Class 4: End-to-end execution — exercised against actual or high-fidelity-simulated embodiments, requires I2 + I4 by construction.

The v1.0 reference implementation will be released under Apache 2.0 and exposed through both a Python binding (for ML integration) and a C binding (for embedded integration). Its source is intended to serve as the canonical-by-construction definition, in the sense that any divergence between the reference implementation and a third-party implementation on a Class 2 test is, by convention, a defect of the third-party implementation rather than of the reference.

Why interface-only at v0.1. Fixing the interface contract while deferring the algorithm is a deliberate sequencing choice. The five invariants above are the load-bearing claims: I1 is what makes conformance testing possible at all; I3 is what makes the compositional algebra of § 4.1 non-vacuous; I2 + I4 are what make the layer safe to deploy in industrial settings. The algorithm satisfying these invariants admits many implementations (inverse-kinematics-first, sampling-based, learned, hybrid); rather than freeze one in v0.1 and inhibit experimentation, the spec freezes the contract and lets v0.1 implementers compete on substrate while remaining conformant. The reference implementation in v1.0 will be one such substrate, not the substrate.

4.3 The Driver Interface

The Driver Interface defines the contract between the Translation Layer and embodiment-specific firmware. It has three components: extended kinematic description, capability dis-

covery, and a runtime action protocol.

Extended kinematic description. RFL extends the URDF/MJCF kinematic-description standard with additional fields required for the spec's purposes:

- Tactile sensor placement and modality (location on the kinematic tree, manifold resolution, native data format)
- Actuation modality (electric, pneumatic, hydraulic, tendon-driven) and its associated dynamic properties (bandwidth, hysteresis, backlash)
- Proprioceptive sensor characterization (joint encoder resolution, force-torque sensor placement and bandwidth)
- Safety-relevant kinematic limits (joint velocity ceilings, end-effector workspace bounds, collision-detection model)
- Calibration metadata (when the embodiment was last calibrated, against which reference, with what residual error)

The extension is additive — a standard URDF parser produces a usable but incomplete embodiment descriptor; an RFL-aware parser produces the full descriptor. This composability with existing tooling is itself a Principle 4 commitment.

Capability discovery. Every RFL-compliant embodiment exposes, at session initialization, a manifest declaring which Skill Layer primitives it can execute, with what fidelity, and under what conditions. The manifest is structured:

```
capability_manifest:
  primitives_supported:
    grasp_pinch:
      fidelity:      "production"          # "experimental" | "stable" | "production"
      success_rate:  0.95                  # empirical on conformance test
      latency_p95:   0.8 seconds
      payload_max:   0.5 kg
      object_size_min: 5 mm
      object_size_max: 80 mm
      tactile_level: "binary_contact"
    grasp_power:
      fidelity:      "production"
    ...
  capabilities_declared:
    {pinch_grasp, power_grasp, lateral_pinch, tactile_sensing,
     force_sensing_3axis, dual_arm: false}
  extensions:
    [rfl-tactile-highres-v0.1]           # opt-in extensions
```

The manifest is the basis for capability negotiation between the VLA and the embodiment. A VLA targeting `grasp_pinch` against an embodiment whose manifest declares `fidelity: experimental` knows to set conservative confidence; an embodiment whose manifest does not declare `pinch_grasp` capability is matched against a different primitive (`grasp_lateral` is the canonical fallback) through the Translation Layer's retargeting algorithm.

Runtime action protocol. At runtime, canonical actions are dispatched to the embodiment through a ROS 2 action server interface. This is a deliberate compatibility commitment with the dominant transport standard (§ 2.1): every RFL-compliant embodiment is, by virtue of compliance, exposed as a set of ROS 2 action servers, one per supported primitive. The action server's goal message carries the canonical action; the feedback messages carry execution telemetry at a declared cadence; the result message carries the outcome plus a structured execution trace. The trace structure is itself standardized — it is consumed by the conformance test suite and by certification tooling, and its standardization is what enables third-party audit of RFL-compliant behavior.

The choice of ROS 2 as the wire protocol does not preclude alternative transports (gRPC, MQTT, DDS-other) for embodiments that do not run ROS 2 internally; the spec defines a wire-format-neutral message structure and provides reference bindings for each major transport. ROS 2 is the default because it is the deployed standard, not because it is privileged in the spec.

4.4 4.4 Worked Example: Cable-and-Connector Insertion

We illustrate the layers together with a worked example drawn from a class of industrial manipulation tasks that motivated the early phase of this work: picking up a flexible cable terminated by a connector and inserting it into a receptacle. The task is structurally non-trivial — it combines compliant-object grasping (the cable is flexible), force-controlled insertion (over-insertion damages pins), and visual servoing (the receptacle pose has tolerance under millimeter scale).

A VLA targeting this task emits a Skill Layer composition:

```
task: insert_cable_into_connector
composition:
  grasp_pinch(target=cable_end_visual, force_budget=3.0N) ;
  reorient(target=connector_alignment_pose) ;
  align(reference=connector_axis, tolerance=2mm) ;
  insert(target=connector_socket, force_budget=8.0N,
    abort_if=force_exceeded) until seated_indicator(force>=4N, depth>=8mm)
```

The Translation Layer expands this composition into a sequence of canonical actions, each parameterized for the target embodiment. For a four-finger tendon-driven hand (e.g., Allegro), `grasp_pinch` resolves to a thumb-and-index opposing grasp with joint trajectory derived from the task-frame target pose; the safety envelope's 3N force budget is expressed as bounded joint torques. For a four-finger direct-drive hand (e.g., LEAP), the same skill resolves to the same opposing geometry but with different torque expressions reflecting the embodiment's actuation properties. For a six-finger pneumatic hand (e.g., Bridgestone TETOTE), the skill resolves to a front-two-finger pinch with the force budget expressed as a pneumatic pressure setpoint (approximately 0.3 MPa for a 3N target force, calibrated through the embodiment descriptor).

Each resolved canonical action is dispatched to the embodiment's ROS 2 action server. The Driver Interface translates the canonical action into motor commands appropriate to the embodiment's actuation. Tactile feedback flows back through the Translation Layer, where it is converted into the TactileManifold representation and made available to the VLA for closed-loop control.

The crucial observation: the VLA's composition is identical across all three embodiments. The VLA does not need to know whether its target is tendon-driven, direct-drive, or pneumatic; it does not need to know whether the hand has four fingers or six. The translation is performed by the layer, not by the model. This is the property that, multiplied across the realistic deployment matrix described in § 1.1, collapses the $N \times M$ integration cost to additive.

A second observation: the composition's insert step uses the Reactive operator (`until_seated_indicator(...)`), which allows the action to terminate cleanly when tactile feedback indicates seating. The Skill Layer's compositional algebra makes this reactive behavior expressible without requiring the VLA to micromanage timing — the embodiment-side implementation of `seated_indicator` is part of the Driver Interface, and its specific implementation varies by embodiment (force-threshold-based for one, deformation-pattern-based for another) but its semantics are uniform.

The example is one of many that the v0.1 design review exercised; the GitHub reference implementation will ship with a curated example library covering each of the seven primitive categories and each of the five composition operators.

4.5 4.5 Conformance and Certification

A specification without a conformance regime is a recommendation, not a standard. RFL ships v0.1 with an open-source conformance test suite that defines, mechanically, what it means for an implementation — VLA or embodiment — to be RFL-compliant.

The suite has four test classes.

Capability conformance. For every primitive an embodiment declares in its capability manifest, the suite executes a battery of test invocations against a calibrated test bench and measures success rate, latency, and safety-envelope adherence. The empirical results become the values populating the manifest; an embodiment's claim of `success_rate: 0.95` is grounded in measurement, not assertion. The test bench specifications are themselves public, so that independent labs can reproduce conformance results.

Composition conformance. The suite executes a fixed library of multi-primitive compositions (sequential, conditional, concurrent, reactive, retry) and verifies that the embodiment's behavior remains within the declared safety envelope of the composition and that the composition's outcome matches the declared semantics. A composition that succeeds on the individual primitives but produces a safety-envelope violation in composition fails the conformance test, regardless of the individual primitive results.

Capability discovery conformance. The suite queries the capability manifest, then attempts primitives that the manifest does and does not declare; the embodiment must succeed on declared primitives at the declared rate and must cleanly reject (not silently fail, not unsafely attempt) primitives it does not declare.

Safety conformance. The suite deliberately invokes primitives at the edge of and beyond their declared safety envelopes (in simulation; certain physical tests are run only on instrumented test articles). The embodiment must abort within the declared response time and must leave the system in a safe state after the abort.

An implementation passing all four classes at a declared spec version is RFL-compliant at that version and may use the RFL™ trademark on packaging and documentation per the trademark policy described in § 7. Implementations passing partial classes are RFL-conformant-subset and may declare which classes they pass, but may not use the trademark.

The conformance regime is what makes the Certification reinforcement loop (§ 5.4 L4) and the Insurance reinforcement loop (§ 5.4 L8) operational. A notified body certifying a deployment against ISO 10218 or ISO 13482 has a defined evaluation surface — the RFL conformance results plus the embodiment's deployment context — rather than having to evaluate the deployment from first principles. An insurer pricing liability against an RFL-compliant deployment has a defined risk profile — the empirical conformance results across the deployed primitives — rather than relying on per-deployment underwriting.

The specification described in this section is what § 5's network-effect machinery operates on, what § 7's governance structure stewards, and what § 9's call to action invites the founding consortium to ratify. It is presented at v0.1 — a deliberate, conservative version number signaling that the architecture is fixed but the detailed schemas, error codes, and extension registry will mature through community review before the v1.0 release planned for late 2027. The companion GitHub repository, available concurrently with this paper, contains the full schema definitions, the reference implementation in Rust with Python and C bindings, the conformance test suite, and the curated example library. Community contributions and consortium proposals enter the spec evolution process described in § 7.

4.6 Open editorial notes

- Word count: $\approx$ 3,650 words (target 3,000–4,000 ✓)
- Status caveat: this is white-paper-grade specification, not RFC-grade. The companion repository must, before v0.1 publication, contain:
 - Complete JSON Schema for every primitive's type signature
 - Complete BNF for the compositional algebra
 - Complete ROS 2 message type definitions
 - URDF/MJCF extension schema
 - Conformance test bench engineering drawings
- Verification flags before final:
 - Confirm the 50-primitive count survives external design review with at least three structurally distinct embodiment partners spanning tendon-driven, direct-drive, and pneumatic actuation classes. The current count is internally consistent but has not yet been red-teamed by independent practitioners
 - The "TactileManifold" abstraction is novel and needs formal mathematical specification in Appendix B; current prose is intuitive but not yet rigorous
 - The Reactive operator's `until` predicate has subtle semantics around concurrent envelope violation that deserve a formal treatment in the spec document, not the white paper
 - Confirm Rust as the reference-implementation language with the prospective CTO; Rust is the right answer but is a commitment worth explicit ratification
- Figures to commission:

- 4.1: Three-layer architecture diagram (Skill / Translation / Driver) with example data flow
- 4.2: Primitive category wheel (seven categories, fifty primitives, with composition operators in the hub)
- 4.3: Capability manifest example (the YAML block in § 4.3 rendered as a structured visualization)
- 4.4: Worked example flow diagram — single VLA composition fanning out to three different embodiments
- 4.5: Conformance regime structure (four test classes, certification path)
- Cross-reference dependencies:
 - § 3 principles each cited at their operationalization point (§ 4.1 for Principle 1 & 3, § 4.2 for Principle 5, § 4.3 for Principle 4, § 4.5 for Principle 3 deep)
 - § 5 inherits the existence of the four reinforcement loops L4–L8 enabled by § 4.5
 - § 7 inherits the trademark policy referenced in § 4.5
 - § 9 inherits the GitHub-repository call to founding consortium

4.7 Coverage check against outline (STRATEGY.md § 15.2)

Outline item	Section	Status
4.1 Skill Layer (□□ ISA)	§ 4.1	✓ (with 50-primitive enumeration + 5 composition operators)
4.2 Translation Layer (□□□□)	§ 4.2	✓ (with canonical action tuple + retargeting algebra)
4.3 Driver Interface (□□ ABI)	§ 4.3	✓ (with extended URDF + capability manifest + ROS 2 protocol)
4.4 Worked example	§ 4.4	✓ (cable-and-connector insertion across three structurally distinct hands spanning direct-drive, tendon-driven, and pneumatic actuation classes)
4.5 Conformance test suite	§ 4.5	✓ (four test classes + trademark gate)

4.8 Revision history

Version	Date	Notes
v1.0	2026-05-29	First English draft. Followed STRATEGY.md § 15.2 outline directly.

5 § 8. Comparative Analysis

A proposal to occupy a structural position is most credibly evaluated by examining the positions that have been occupied before and the ones that have remained empty despite attempts. This section places RFL alongside three historical coordination layers chosen for their structural relevance — ARM, CUDA, and USB — and the contemporary cohort that adjacent to RFL but does not occupy its position. The aim is not to claim RFL will replay any of these histories, but to identify the structural dimensions on which RFL must satisfy the conditions that prior successes satisfied and avoid the conditions that prior failures encountered.

5.1 8.1 The Framework

Coordination layers vary along three dimensions that matter for long-run defensibility. The first is technical position: the layer of the stack at which the abstraction is drawn, the side or sides of the market it abstracts, and the kind of object it standardizes (instruction, message, data format, action). The second is governance form: who owns the specification, who controls its evolution, how participation is regulated. The third is constitutional discipline: which commitments the steward refuses to abandon under commercial pressure, and which structural mechanisms enforce that refusal.

Each historical case occupies a distinct combination on these dimensions, and the combinations have proved more or less durable. The analysis below maps RFL against three of them in turn.

5.2 8.2 ARM: The Structural Correspondence

ARM is the closest structural analog and the case from which RFL's design pattern is most directly derived. § 3.3 stated the eight-element correspondence between ARM ISA design choices and RFL specification choices. § 8 emphasizes the dimensions on which ARM's history offers operational lessons for RFL's trajectory.

Three specific moments in ARM's history bear directly on choices RFL faces. First, ARM's decision in 1990 to spin out of Acorn as an independent entity — sacrificing the captive customer that funded its research to gain the neutrality required for cross-vendor adoption. The parallel for RFL is the choice to organize as a Linux Foundation subsidiary rather than as a wholly-owned project of any Founding Member, and to bind the Foundation's revenue model in ways that prevent vertical capture. The structural lesson: neutrality has a price paid up-front and a value recovered over decades.

Second, ARM's near-death experience in the desktop market from 2000–2005, when Intel's superior process technology threatened to make ARM irrelevant outside embedded niches. ARM's survival came from doubling down on the mobile market — accepting that desktop dominance was unwinnable and pivoting toward the segment where its energy-efficiency advantage compounded into a structural lead. The parallel for RFL is the imperative to choose the segment where the N×M crisis is most acute and the alternative (vertical integration) is most painful — likely the industrial manipulation and humanoid-services segments where the embodiment diversity is highest and no single VLA provider has yet locked in dominant share. The structural lesson: coordination layers win the segment where vertical integration is least tolerable, not

necessarily the segment of largest absolute size.

Third, ARM's response to the v8 → v9 architectural transition (2018–2022), which preserved binary compatibility for the v8 instruction base while adding new architectural features. ARM's discipline in not breaking compatibility, even when commercial parties advocated for it, protected the multi-decade application base on which its position rested. RFL's forward-compatibility principle (§ 3.2 Principle 5) is the descendant of this discipline, and § 7.4's evolution procedure is its operational instantiation.

ARM's history demonstrates that the structural position RFL seeks is occupiable, that its defensibility is durable, and that the commercial pressures against neutrality are real and recurring. ARM's history does not, however, demonstrate that physical-AI specifically will follow the same trajectory; that question is settled by execution.

5.3 8.3 CUDA: The Cautionary Inverse

CUDA occupies the cautionary inverse of ARM's position. It is a coordination layer in a technical sense — application code targeting CUDA runs across NVIDIA's GPU product line and benefits from the network of optimized libraries (cuBLAS, cuDNN, TensorRT) — but it is owned and operated by a single vendor whose primary commercial interest is selling the underlying hardware. The result is a layer of extraordinary technical sophistication that has nevertheless attracted, over the past three years, an aggressive industry-wide effort to escape it (the Triton language from OpenAI, the MAX platform from Modular, Intel's oneAPI, AMD's ROCm, Apple's Metal Performance Shaders, and the emerging cohort of inference compilers targeting non-NVIDIA accelerators).

The CUDA case shows what RFL must not become. A coordination layer whose steward also competes in adjacent markets generates a structural incentive for adopters to seek alternatives the moment the alternatives become technically viable. NVIDIA's commercial response — adding more value at the CUDA layer faster than the alternatives can erode adoption — is sustainable while NVIDIA's technical lead in silicon is unassailable, but it is not a structural defense; it is a continuously running engineering campaign. RFL's neutrality commitment (§ 7.6 commitment 1) is the structural alternative to this campaign.

The CUDA case also shows what is at stake in the choice. Had NVIDIA chosen to spin CUDA out into a neutral foundation in 2014, when its position was strong enough to dictate terms, the GPU computing ecosystem would today have a single dominant abstraction layer rather than the fragmenting cohort of alternatives that the captive ownership produced. The opportunity that NVIDIA declined is the opportunity RFL is positioned to take in physical AI before the equivalent verticalization ossifies.

5.4 8.4 USB: An Alternative Governance Model

USB occupies a third position relevant to RFL's design: it is a consortium-governed standard (the USB Implementers Forum, a non-profit), narrower in scope than either ARM or CUDA, but with notably broader cross-vendor adoption than either. The USB-IF model — a member-funded consortium that owns the specification, certifies compliant implementations, and licenses the trademark — is the closest existing governance model to the RFL Foundation form

described in § 7.

Three features of USB-IF's structure are directly adopted by RFL. First, the trademark gate: USB devices cannot legally display the USB logo without passing certification, and this commercial-grade legibility of conformance has proved more effective at enforcing compatibility than any technical mechanism. RFL adopts the same pattern (§ 7.5). Second, the working-group structure: USB-IF's technical work is organized into focused working groups with rotating chairs from member organizations, which prevents single-organization dominance and ensures multiple perspectives shape each spec revision. RFL's four working groups (§ 7.3) follow the same template. Third, the modular extension mechanism: USB has evolved from 1.0 through 4.0 over thirty years by adding capabilities without breaking compatibility, governed by the same body throughout. RFL's extension registry (§ 6.1) is structurally similar.

Two features of USB-IF's structure are deliberately not adopted. USB-IF's membership dues are dominated by a small number of large hardware companies, which has produced occasional concerns about asymmetric influence over spec direction; RFL's three-tier membership structure (§ 7.2) is designed to dilute this risk. And USB-IF has no equivalent of RFL's constitutional commitments (§ 7.6); the absence has not been problematic in USB's history but is a deliberate strengthening in a domain where the long-run governance pressures are more intense.

5.5 8.5 The Contemporary Cohort

The contemporary efforts — ROS 2, LeRobot, Open X-Embodiment, and the NVIDIA Isaac/GR00T stack — were treated in § 2 as adjacent efforts that do not occupy the RFL position. From the comparative-analysis perspective, the additional observation worth making is that each occupies a different structural position from the others, and the differences themselves are evidence that the physical-AI coordination problem has been recognized but not resolved.

ROS 2 is governance-strong (Open Source Robotics Foundation, multi-stakeholder) but technical-position-narrow (transport layer, not action semantics). LeRobot is technical-position-broader but governance-weak (Hugging Face captive). Open X-Embodiment is technical-position-narrow (training data, not runtime) and governance-asymmetric (Google-dominated). NVIDIA Isaac is technical-position-broadest of the four (full stack) but governance-foreclosed (vertically integrated). Each pair of strengths is missing the complementary strength that the structural position requires. RFL is designed to combine ROS 2's governance discipline with Isaac's technical scope, neither of which exists in isolation.

5.6 8.6 The Position That Emerges

Set the analyses together and the position RFL is designed to occupy resolves to a specific combination:

Dimension	ARM	CUDA	USB	RFL
Technical layer	ISA	Compute API	Bus	Action semantics + embodiment translation

Dimension	ARM	CUDA	USB	RFL
Sides abstracted	OS ↔ silicon	App ↔ GPU	Host ↔ device	VLA ↔ embodiment
Governance form	Single company (neutral)	Single vendor	Consortium	LF-hosted foundation
Constitutional binding	High (de facto)	None	Medium	High (de jure)
Verifiable safety	n/a	n/a	n/a	Yes (§ 4.5)
Trademark gate	License-mediated	Vendor-controlled	Certification-gated	Certification-gated
Cross-vendor adoption	Very high	Declining	Very high	(target)

The combination is novel along two axes. First, RFL pairs ARM-class technical depth with USB-class consortium governance — a combination neither historical case achieved (ARM was governance-light, USB was technically narrow). Second, RFL introduces verifiable safety semantics as a first-class architectural concern, which none of the historical cases needed to address because none of them mediated physical action.

The empty position in the historical landscape is therefore not a contingent gap but a structural one: it exists because no prior coordination problem combined the cross-vendor abstraction need with the physical-safety need at the layer where the abstraction is drawn. The physical-AI coordination problem is the first to do so, and the abstraction layer that resolves it must combine governance and verification disciplines that have not previously been combined. The Foundation, the specification, and the principles described in §§ 3–7 are the design that combines them.

5.7 Open editorial notes

- Word count: $\approx$ 1,500 words (target 1,200–1,500 ✓)
- Tone: comparative-analytical, not promotional. The position should resolve from the structural comparison; the reader should reach the conclusion before § 8.6 spells it out
- Verification flags before final:
 - ARM 1990 Acorn spin-out date: confirm
 - ARM v8→v9 transition window: confirm 2018–2022 framing
 - “\$X under-counts CUDA escape effort” framing not yet quantified; if cited, source needed (state-of-foundations report, e.g.)
 - USB-IF membership-fee asymmetry observation: cite a credible source if claim retained
- Figures to commission:
 - 8.1: Three-dimensional positioning chart (technical position $\times$ governance form $\times$ constitutional discipline) with the four cases plotted
 - 8.2: The table in § 8.6 rendered as a side-by-side capability / governance comparison
 - (optional) 8.3: Timeline overlay showing ARM, CUDA, USB founding moments and RFL's projected position on the same axis
- Cross-reference dependencies:

- § 3.3 ARM isomorphism is referenced; § 8.2 deepens the historical lessons rather than duplicating
- § 5.6 comparison table is the precursor to § 8.6's expanded analysis
- § 7 governance structure is the operational expression of the § 8.6 position

5.8 Coverage check against outline (STRATEGY.md § 15.2)

Outline item	Section	Status
8. Comparative Analysis (2 □□□) — RFL vs ARM vs ROS vs CUDA vs HF vs Open X-E	§§ 8.1–8.6 (framework + ARM + CUDA + USB + contemporary cohort + synthesis)	✓

5.9 Revision history

Version	Date	Notes
v1.0	2026-05-29	First English draft. ARM, CUDA, USB as historical cases; contemporary cohort consolidated into § 8.5.

6 § 6. Implementation Roadmap

7 § 7. Conclusion

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abandoned” the Gen 3 rolling-contact design after sim-to-real failures, and the V3 hands at the summer 2026 reveal will use a newer, undisclosed design.

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8.9 I. Industry analyst and market estimates (§ 1) — DELETED IN v1.3

Entries 52, 53, and 54 (placeholder slots for PitchBook / CB Insights VC financing aggregates, industry-analyst “suppressed market value” estimates, and 2026 robotics-AI engineering compensation surveys) were deleted in v1.3 (2026-05-30). The quantitative claims those placeholders would have supported are resolved in the body via Path A (soften) — see appendix_A_data_sources_en.md § A.4–A.6 for the full softening record. No body citation depends on these entries; deletion is safe.

9 Appendices

10 Appendix A. TactileManifold: Formal Specification

The TactileManifold is the RFL Translation Layer’s canonical representation of tactile state. Its design satisfies a constraint that is unmet by any existing tactile-sensor format: it must express the observations of a binary four-channel contact sensor, a three-axis force-torque sensor, a high-resolution GelSight-class visuotactile sensor, and as-yet-undesigned tactile modalities in a single uniform structure that admits capability negotiation. This appendix gives the formal

definition, the resolution hierarchy that enables negotiation, and the realizations for the principal sensor classes in 2026 commercial use.

10.1 B.1 Motivation and Design Constraints

The body section establishes the structural requirement: a VLA expressing a high-level skill (e.g., `grasp_pinch`) must be able to specify a tactile target without committing to a particular sensor format. An embodiment receiving the target must be able to report its observations against it. Different embodiments with different sensors must produce comparable observations. The abstraction therefore satisfies three constraints simultaneously:

1. Format independence: a sensor's native output format does not appear in the manifold definition. Sensor-specific encoding is performed by the embodiment's Driver Interface during translation.
2. Resolution heterogeneity: a low-resolution sensor and a high-resolution sensor must express observations in the same abstraction, with the resolution difference captured as a parameter rather than as a different structure.
3. Graceful degradation: if a VLA requests a feature the embodiment cannot supply, the negotiation must fail cleanly with a structured error, not silently produce a degraded observation that the VLA misinterprets.

The third constraint is the load-bearing one. Many existing tactile-fusion approaches produce a single output that hides the underlying resolution differences, with the consequence that a VLA trained on high-resolution data degrades unpredictably when deployed against a low-resolution sensor. The TactileManifold abstraction is designed to make the resolution explicit at every layer.

10.2 B.2 Formal Definition

A TactileManifold is a tuple

$$\mathcal{M} = (S, \mathcal{F}, R, t)$$

where:

- $S \subseteq \mathbb{R}^3$ is a spatial domain, a (possibly discrete) subset of the embodiment's end-effector surface, equipped with a topology induced by the surface metric. For dense sensors, S is a two-dimensional manifold patch; for discrete sensors, S is a finite set of points; both are admitted by the definition.
- $\mathcal{F} : S \rightarrow \mathbb{V}$ is a feature field, assigning to each point in S a vector of contact features drawn from a feature vocabulary $\mathbb{V}$. The vocabulary is defined below and is the structure that admits cross-modality comparison.
- $R = (r_s, r_t, r_f)$ is a resolution triple, where r_s is the spatial density (points per unit area, or "discrete" for sparse sensors), r_t is the temporal sample rate (Hz), and $r_f : \mathbb{V} \rightarrow \mathbb{N}$ is a per-feature precision function (bits of effective dynamic range per feature channel).
- $t \in \mathbb{R}_{\geq 0}$ is the timestamp of the manifold instance.

The feature vocabulary $\mathbb{V}$ is defined as a fixed, extensible set of named feature channels:

$$\mathbb{V} = \{\text{contact}, \text{normal_force}, \text{shear}_x, \text{shear}_y, \text{slip}, \text{vibration}, \text{temperature}, \text{compliance}, \dots\}$$

Each named channel has a fixed semantic interpretation specified in the RFL spec document. The set is extensible through the extension registry (§ 6.1), so future sensor modalities can register new channels without breaking the core specification.

A partial manifold is an instance in which $\mathcal{F}$ is defined only over a subset of the vocabulary. The spec language requires the embodiment's capability manifest to declare which channels it supplies, so the partial nature is contractual and verifiable rather than silent.

10.3 B.3 The Resolution Hierarchy

The resolution triple R admits a partial order $\succeq$:

$$R_1 \succeq R_2 \iff r_{s,1} \geq r_{s,2} \wedge r_{t,1} \geq r_{t,2} \wedge r_{f,1}(c) \geq r_{f,2}(c) \forall c \in \mathbb{V}$$

This partial order is the structure that supports capability negotiation. When a VLA emits a Skill Layer invocation with a tactile target specified at resolution R_{req} , the Translation Layer queries the embodiment's capability manifest for the native resolution R_{native} . If $R_{\text{native}} \succeq R_{\text{req}}$, the invocation proceeds; the embodiment supplies a manifold instance at R_{native} (or, optionally, downsamples to R_{req} to reduce bandwidth). If $R_{\text{native}} \not\succeq R_{\text{req}}$, the negotiation fails with a structured error identifying the dimension(s) on which the requested resolution exceeds the embodiment's capability.

The asymmetric nature of the partial order is intentional. A VLA that requests low resolution and receives high resolution can use the additional information or discard it without error. A VLA that requests high resolution and receives low resolution cannot infer the missing information; the failure must be visible. The Translation Layer enforces this by treating the resolution declaration as a contract rather than a hint.

10.4 B.4 Realizations from Specific Sensor Classes

Five sensor classes in 2026 commercial use are realized as follows. The mapping shown for each is the canonical one defined in the RFL Reference Implementation; embodiment vendors may register alternative mappings through the extension registry if their sensor admits more refined representation.

Binary contact sensors (four-channel discrete). Used in many low-cost grippers. Mapping: S is a discrete four-point set corresponding to fingertip locations; $\mathcal{F}(s) = \{\text{contact}: \{0, 1\}\}$ for each $s \in S$; $R = (\text{discrete}, 100 \text{ Hz}, r_f(\text{contact}) = 1)$. Only the contact channel is populated; the remaining vocabulary is undefined.

Three-axis force-torque sensors at the wrist. Used in most collaborative arms. Mapping: S is a single-point set at the force-sensor location; $\mathcal{F}(s) = \{\text{normal_force}, \text{shear}_x, \text{shear}_y\}$ as three

scalars; $R = (\text{discrete}, 1000 \text{ Hz}, r_f(\cdot) = 16)$. The three force channels are populated; spatial channels are absent because the sensor is point-localized.

Distributed force-sensor arrays (XELA-class). Used in high-end multi-finger hands. Mapping: S is a discrete set of sensor locations (typically 8–24 per fingertip); $\mathcal{F}(s) = \{\text{normal_force}, \text{shear}_x, \text{shear}_y\}$ at each location; $R = (\text{discrete}_n, 100 \text{ Hz}, r_f(\cdot) = 14)$ where n is the per-fingertip array size.

Visuotactile sensors (GelSight, DIGIT classes). Used in research-grade and emerging industrial deployments. Mapping: S is a dense two-dimensional patch (e.g., 320×240 pixels mapped to the elastomer-gel surface); $\mathcal{F}(s) = \{\text{normal_force}, \text{shear}_x, \text{shear}_y, \text{slip}\}$ derived from the gel-deformation image; $R = (\text{dense}, 30 \text{ Hz}, r_f(\cdot) = 8)$.

Pneumatic pressure sensors (as in McKibben and bellows-style pneumatic hands). Used in pneumatic-hand embodiments. Mapping: S is a discrete set of finger-chamber locations (e.g., six chambers in a six-finger pneumatic hand); $\mathcal{F}(s) = \{\text{normal_force derived from chamber pressure}\}$; $R = (\text{discrete}, 50 \text{ Hz}, r_f(\text{normal_force}) = 12)$. The pressure-to-force conversion is calibrated through the embodiment descriptor's calibration metadata.

This list is not exhaustive; the design intent is that any tactile sensor whose output can be characterized by a spatial distribution of feature values can be expressed in TactileManifold form. The capability negotiation mechanism then handles the heterogeneity at runtime.

10.5 B.5 Capability Negotiation Semantics

For a Skill Layer primitive whose tactile target is $\mathcal{M}_{\text{req}} = (S_{\text{req}}, \mathcal{F}_{\text{req}}, R_{\text{req}}, \cdot)$, the negotiation against an embodiment with native capability $\mathcal{M}_{\text{native}} = (S_{\text{native}}, \mathcal{F}_{\text{native}}, R_{\text{native}}, \cdot)$ proceeds in three steps:

1. Feature coverage: every channel in $\mathcal{F}_{\text{req}}$ must be present in $\mathcal{F}_{\text{native}}$. If not, negotiation fails with `UnsupportedFeatureError(channels=...)`.
2. Spatial coverage: S_{req} must be approximable by points in S_{native} within a declared spatial tolerance. If not, negotiation fails with `SpatialCoverageError(uncovered_region=...)`.
3. Resolution adequacy: $R_{\text{native}} \succeq R_{\text{req}}$ as defined in B.3. If not, negotiation fails with `InsufficientResolutionError(dimensions=...)`.

The Driver Interface returns the structured error rather than silently substituting a fallback. The VLA's response to negotiation failure is its own concern; the Translation Layer's responsibility is to make the failure visible and structured.

The negotiation produces, on success, an adapted manifold $\mathcal{M}_{\text{adapted}}$ which is the embodiment-native manifold downsampled to R_{req} (if downsampling is requested) or supplied at R_{native} (the default). The VLA receives observations against $\mathcal{M}_{\text{adapted}}$, with a metadata field indicating which resolution was used.

10.6 B.6 Open Questions

Two questions are left to future spec revisions.

First, the formal treatment of deformable-object tactile interaction — where the contact surface itself deforms during interaction and the spatial domain S should be time-varying — is sketched but not specified. The current spec accepts S as time-invariant during a single primitive invocation; future revisions will need to address adaptive spatial domains for cloth, soft-object, and slip-rich manipulations.

Second, the relationship between TactileManifold and the proprioceptive state captured separately in the canonical action representation (§ 4.2) admits some overlap (force at the wrist appears in both). The current spec treats them as separate channels; a future revision may unify them under a richer end-effector-state abstraction.

These are flagged for v2.0 of the specification, consistent with the extension-registry policy (§ 6.1) by which the v0.1 / v1.0 core stays small and well-defined while non-trivial new structure migrates through extensions before promotion.

11 Appendix B. Supermodularity of Bilateral Translation

This appendix formalizes one proposition stated informally in § 4: that the value of a translation layer with high-fidelity models of both sides (VLA models and embodiments) strictly exceeds the value attainable with high-fidelity modeling of either side alone. The argument is mathematically straightforward once the setup is precise; the appendix exists primarily to make the setup precise.

11.1 C.1 Notation and Setup

Let $\mathcal{V}$ denote the (countable) set of Vision-Language-Action models in active use, indexed by $v \in \mathcal{V}$, and let $\mathcal{E}$ denote the (countable) set of robotic embodiments, indexed by $e \in \mathcal{E}$. A deployment instance is a pair $(v, e) \in \mathcal{V} \times \mathcal{E}$ paired with an environment context $\xi \in \Xi$ and a task specification $\tau \in \mathcal{T}$.

For each deployment instance, the RFL Translation Layer realizes a function

$$T : \mathcal{A}_v \times \Theta_v \times \Phi_e \rightarrow \mathcal{A}_e$$

where $\mathcal{A}_v$ is the action space output by VLA v , Θ_v is the information held by the Translation Layer about v 's semantics (its action distribution, its calibration, its uncertainty model), Φ_e is the information held about embodiment e (its kinematic structure, its actuation modality, its safety envelope), and $\mathcal{A}_e$ is the embodiment-specific command space.

Define the translation value $V(\theta, \phi)$ as the expected utility of the translation across a representative distribution over (ξ, τ) :

$$V(\theta, \phi) = \mathbb{E}_{(\xi, \tau)} [U(T(a; \theta, \phi)) \mid a \sim P_v(\cdot \mid \xi, \tau)]$$

where $\theta \in \Theta_v$ is the VLA-side information level, $\phi \in \Phi_e$ is the embodiment-side information level, a is the action sampled from the VLA, and U is a scalar utility composing task success, safety-envelope adherence, and execution efficiency.

We treat θ and ϕ as scalar information levels (e.g., empirical KL-divergence between the Translation Layer's belief over the relevant distribution and the true distribution). The vector-valued generalization is straightforward and is omitted to keep notation legible.

11.2 C.2 Supermodularity of Translation Value

Proposition 1 (Supermodularity of V). Under regularity conditions on U and the underlying distributions stated below, the value function $V : \Theta_v \times \Phi_e \rightarrow \mathbb{R}$ is supermodular in (θ, ϕ) . Equivalently, for any $\theta_L < \theta_H$ and $\phi_L < \phi_H$:

$$V(\theta_H, \phi_H) - V(\theta_H, \phi_L) \geq V(\theta_L, \phi_H) - V(\theta_L, \phi_L) \quad (\text{C.1})$$

with strict inequality whenever both $\theta_H - \theta_L > 0$ and $\phi_H - \phi_L > 0$ have non-trivial impact on T 's output.

Regularity conditions.

1. U is differentiable and bounded on the relevant compact set.
2. The map T is jointly continuous in (a, θ, ϕ) and differentiable almost everywhere.
3. The distributions P_v and the embodiment-side noise distributions are not degenerate (i.e., the bilateral information actually matters; neither side is fully observable from the other).

Proof sketch. The argument follows Topkis (1998) Theorem 2.6.1, adapted to the translation setting.

Step 1: Cross-partial sign. For V twice continuously differentiable in (θ, ϕ) , supermodularity is equivalent to $\partial^2 V / \partial \theta \partial \phi \geq 0$ everywhere on the domain (Topkis 1998, Lemma 2.6.2). We compute this cross-partial.

Holding a fixed, the contribution to V at point (θ, ϕ) from any single a is $u(T(a; \theta, \phi))$ where $u = U$ composed with the post-translation execution model. The cross-partial of $u(T(a; \theta, \phi))$ in (θ, ϕ) is:

$$\frac{\partial^2 u(T)}{\partial \theta \partial \phi} = u''(T) \cdot \frac{\partial T}{\partial \theta} \cdot \frac{\partial T}{\partial \phi} + u'(T) \cdot \frac{\partial^2 T}{\partial \theta \partial \phi}$$

The first term is non-negative whenever u is concave (a standard assumption for utility) and the two single-side partial derivatives have the same sign — which they do, since improving information on either side improves translation accuracy in the same direction (toward the true optimal command).

The second term — the cross-partial of T itself — is non-negative under the structural assumption that improvements in VLA-side information and embodiment-side information combine constructively in the translation. This is the substantive claim, and it follows from the form of the optimal translation: under joint information (θ_H, ϕ_H) , the translator can map VLA outputs

to embodiment commands that exploit the joint distribution; under either piece alone, the translator must marginalize over the missing dimension, which introduces irreducible variance. The combination strictly reduces variance and thereby strictly improves the expected utility under concave u .

Step 2: Integration over a . Taking the expectation over $a \sim P_v$ preserves the sign of the cross-partial pointwise (by linearity of expectation), giving $\partial^2 V / \partial \theta \partial \phi \geq 0$ on the domain, with strict inequality wherever the structural assumption in Step 1 is non-degenerate.

Step 3: Supermodularity inequality. Integration of the cross-partial twice (once in θ , once in ϕ) over the rectangle $[\theta_L, \theta_H] \times [\phi_L, \phi_H]$ yields the supermodularity inequality (C.1). ■

Connection to Crémer-McLean (1985, 1988). The original Crémer-McLean result, in the mechanism design setting, establishes that a monopolist with knowledge of the correlation structure between bidders' types can extract full surplus, while a monopolist with knowledge of only the marginal distributions cannot. The translation setting is structurally analogous: the Translation Layer is a "mechanism" whose payoff depends on the joint structure of two distributions (VLA action distribution and embodiment response distribution), and bilateral information enables strict outcome dominance over unilateral information. The supermodularity inequality is the Topkis-form expression of the same complementarity.

Empirical confirmation. The proposition's empirical consequence — that joint training on multi-VLA $\times$ multi-embodiment data outperforms either unilateral training — is observed in published results. Physical Intelligence (2024) reports super-linear improvements when π_0 is fine-tuned on multi-embodiment data relative to single-embodiment fine-tunes of the same parameter count. The Open X-Embodiment results (Padalkar, Pooley, Jain, et al., 2023) show analogous transfer gains across nine embodiments. These results are consistent with the proposition but do not, in themselves, prove it; the proposition's value is that it predicts the gains should continue to scale with the joint information, which is testable.

11.3 B.3 Out of Scope

The body of an earlier draft of this work proposed a "Master Equation" characterizing long-run RFL value as a multiplicative-with-constraints functional. That material has been extracted to a separate position paper because the underlying form is illustrative rather than predictive — it depends on a free scaling coefficient calibrated only against historical platform analogs — and does not constitute a specification claim. The appendix retains only the supermodularity result, which is a structural property of the translation problem and therefore properly part of the specification's formal basis.

End of RFL Specification v0.1